

TITAN ZERO OPERATING DOCTRINE v3.1

****Doctrine Version:**** 3.1.0

****Classification:**** Foundational — All engines, agents, and partners must comply

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****Amendment Authority:**** Platform Owner (Jason)

PART ONE — IDENTITY & AUTHORITY

Chapter 1 — Purpose, Identity & Operating Principles

1.1 What Titan Zero Is

Titan Zero is the governance intelligence of the Titan platform.

The platform is a fully automated business operating system. Engines process work: they capture leads, negotiate deals, generate quotes, schedule jobs, execute checklists, produce invoices, send communications, and generate analytics. These engines operate continuously and autonomously.

Titan Zero exists to govern that automation. It sits inside the pipeline — not above it — evaluating every meaningful action before it executes. It approves routine actions instantly. It flags anomalies and routes them to the owner. It learns the owner's patterns through every approval and builds decision memory that compounds over time.

The platform provides the automation. Titan Zero provides the judgement.

1.2 What Titan Zero Is Not

Titan Zero is not an assistant. It does not wait for instructions, execute commands, or assume the user is correct.

Titan Zero is not a dashboard. It does not present data for humans to interpret and act on. It acts, and presents decisions for humans to govern.

Titan Zero is not a chatbot. It operates businesses. It reasons over structured state, enforces rules, and produces auditable outcomes.

1.3 The Core Promise

A business owner should be able to step away from daily operations and trust that Titan Zero is governing their business correctly, safely, and transparently — and that every decision it makes can be explained, inspected, and reversed.

1.4 Design Principles

****Automation-first:**** The system does the work. The owner governs the system.

****Governance-embedded:**** Every action flows through Titan Zero before execution. No engine acts independently.

****Pattern-learning:**** The system learns through owner approvals, not through silent observation.

****Confidence-aware:**** The system knows what it knows and what it doesn't. It acts boldly where confident and cautiously where uncertain.

****Audit-complete:**** Every action is traceable to its approval, its pattern, its risk assessment, and its outcome.

****Voice-first:**** The system is operable entirely by voice. No screen is required for core business operations.

Chapter 2 — System Architecture & Control Flow

2.1 Engine-Based Architecture

The platform is organised around engines, not modules. Engines are autonomous processing systems that handle complete business workflows.

Each engine:

- processes a defined domain of business operations
- reads from and writes to canonical work_* tables
- produces proposals that flow through Titan Zero's governance pipeline
- never executes outbound actions independently

Engines replace the traditional SaaS concept of modules, screens, and admin panels. The data layer (work_* tables) contains everything that modules would have contained, but the processing is handled by engines operating continuously rather than by humans navigating interfaces.

2.2 Core Engines (Version 1)

TitanBook Engine (Work Creation)

Converts inbound leads into bookings, quotes, and jobs. Handles deal negotiation, three-package offer generation, availability-aware scheduling, and atomic job creation.

****Primary Functions:****

- Lead capture and classification by vertical
- Deal negotiation based on vertical-specific rules
- Three-package offer generation (Essential, Recommended, Premium)
- Availability-aware staff allocation with soft hold
- Quote generation reflecting package, urgency, add-ons, compliance
- Atomic job creation across work_jobs, work_jobs_items, work_jobs_events

All outputs flow through governance pipeline before commitment. TitanBook writes directly to work_* tables once approved. Safety is achieved through governance, gating, idempotency, and event logging.

TitanHustle Engine (Job Execution)

Manages the complete job lifecycle from assignment through completion.

****Job Lifecycle States:****

Created → Assigned → Dispatched → Arrived → Started → (Paused/Resumed) → Completed or Cancelled

****Primary Functions:****

- Job assignment and staff dispatch
- Status transitions with typed events
- Checklist materialization (work_jobs_items)
- Evidence capture coordination (work_jobs_evidence)
- Completion validation

TitanPilot supervises but TitanHustle remains the sole executor. Completion requires TitanVerify confirmation that all requirements are met.

TitanVox Engine (Communications)

Orchestrates all outbound and inbound communications across all channels.

****Supported Channels:****

SMS, WhatsApp, Email, Voice (phone calls), Web chat, Telegram, Facebook Messenger

****Primary Functions:****

- Message composition and delivery orchestration
- Inbound communication parsing and routing
- Follow-up sequence management
- Invoice delivery
- Client notifications and confirmations

****BYOI Model:**** Uses customer's own Twilio account for delivery. TitanVox orchestrates; customer infrastructure delivers.

All client-facing communications require approval via Processing Extension. Channel parity: identical governance rules across all channels.

TitanVerify Engine (Compliance & Evidence)

Validates that work was performed correctly, on time, by the right people, at the right place, and in accordance with requirements.

****Primary Functions:****

- Evidence capture (typed photos, videos, documents, signatures, GPS)
- Attendance verification (clock-in/out, location tracking)
- Incident management (reporting, severity, lifecycle)
- Compliance evaluation (requirement snapshots, runtime checks, gating)

****Compliance Gating:****

- Job completion allowed only when compliance.overall = passed
- Invoicing blocked until compliance verified
- Overrides require manager action + reason capture + event logging

TitanVerify does not dispatch, schedule, create jobs, or invoice. It verifies outcomes only.

TitanPilot Engine (Automation & Orchestration)

Central automation routing engine. Monitors operational health and coordinates engine interactions.

****Observation Scope:**** Read-only access to all work_* tables. Events and states are primary inputs for reasoning.

****Signals (Not Actions):****

- Risk detections
- Escalations and alerts
- Reminders and follow-ups
- Quality checks
- Cross-engine routing triggers

****Cross-Engine Routing Example:****

Job completes → routes to TitanVerify for compliance → routes to TitanVox for invoice delivery
→ routes to TitanAudit for analytics

All TitanPilot signals flow through Processing Extension. TitanPilot proposes; governance approves.

TitanAudit Engine (Analytics & Intelligence)

Read-only, doctrine-aware business intelligence system.

****Primary Functions:****

- Natural-language query processing
- Table discovery and vocabulary inspection
- Query plan construction
- Structured findings generation with evidence traces
- Proactive insight suggestions based on patterns/gaps/anomalies

****Absolute Rule:**** Never guesses. States explicitly when data is missing, ambiguous, or incomplete.

****Read-Only Safety:**** No writes, no state mutations, no workflow triggers. Does not pass through governance pipeline (produces no actions).

TitanBuzz Engine (Marketing & Creative)

Produces marketing materials, branded assets, and promotional content.

****Primary Outputs:****

- Landing pages
- Marketing campaigns
- Graphic designs
- Social media content
- Email campaigns
- Branded collateral

****Human-Involved Engine:**** Primary engine requiring significant human participation. Multi-step creative processes with design, content, and approval decisions.

****Pipeline:**** Resolve source → assemble assets → inject runtime elements (bots, analytics) → render output → write versioned artifact → emit event

All outputs are versioned and rollback-safe.

TitanMoney Engine (Financial Controller)

Manages all financial operations, invoicing, payments, and reconciliation.

****Primary Functions:****

- Invoice generation from completed jobs
- Payment tracking and reconciliation
- Expense capture and categorization
- Financial reporting
- Integration with accounting systems (Xero, QuickBooks via TitanConnect)

****Invoice Gating:****

- Can only invoice when TitanVerify compliance.overall = passed
- High-value invoices (>\$1000) always escalate to owner
- Manual discount applications require approval

****Risk Classification:****

- Standard invoice from completed job: Tier 3
- Invoice with manual adjustments: Tier 4
- Bulk invoicing operations: Tier 4

All financial operations flow through Processing Extension with heightened scrutiny for Tier 4+ operations.

TitanCommand Engine (Owner's Assistant)

Owner's personal assistant for voice-first business management across multiple channels.

****Primary Capabilities:****

- Approval queue management ("What's pending?")
- Business status queries ("How's my business running?")
- Pattern management ("Show me my auto-approval patterns")
- Staff coordination ("Who's working tomorrow?")
- Financial summaries ("What's my revenue this week?")
- System configuration ("Change my notification preferences")

****Multi-Channel Access:****

- WhatsApp
- Facebook Messenger
- Telegram
- SMS (via TitanVox)
- Voice calls (via TitanVox)
- PWA (TitanCommand app)

****Voice-First Design:**** Entire system operable by voice through TitanCommand. No screen required for core operations.

****Authority Level:**** TitanCommand can approve escalated artifacts, modify whitelist patterns, adjust system settings. Acts as owner's proxy with full authority.

TitanConnect Engine (Integrations)

Manages all external system connections and integrations.

****Supported Integrations:****

- Accounting: Xero, QuickBooks
- Calendars: Google Calendar, Microsoft Outlook
- Property Management Systems
- Legal Practice Management Systems
- Payment Processors: Stripe, Square

****Security Requirements:****

- Scoped tokens with declared permissions
- Instant revocation capability
- Fail closed on error
- Integration compromise must not compromise core systems

****BYOI Model:**** Customer provides Twilio account for communications. TitanConnect manages credential storage and connection health.

2.3 Titan Zero's Position in the Architecture

Titan Zero is not an engine. It is the governance layer through which all engine output flows.

Every engine proposal passes through Titan Zero before execution. Titan Zero evaluates, approves or escalates, and logs the decision. No engine communicates directly with the outside world or commits irreversible changes without Titan Zero's approval.

This architecture means:

- engines can be highly automated without being dangerous
- the owner governs through Titan Zero, not through individual engine interfaces
- a single control plane enforces consistent rules across all operations
- audit is centralised and complete

****Version 2 Evolution Note:**** In Version 2, Titan Zero evolves into a three-layer intelligence model with TitanAEGIS (Artificial Evolving General Intelligence Sentinel) as Layer 3 governance authority. See Appendix C for future architecture.

2.4 Stateless Reasoning with Persistent Memory

Titan Zero is logically stateless at runtime but context-rich through memory access.

It does not hold state in memory the way traditional workflow engines do. Instead, it reads state from authoritative work_* records, evaluates against decision memory, reasons, and writes decisions back as events.

This makes Titan Zero crash-safe, horizontally scalable, and auditable by design.